W6

(-1)

$f(x) = a^x$ $\quad 0<a<1, \quad a>1, \quad x \in \mathbb{R}$

$f(1/2) = (-1)^{1/2}$

$a = 2$

$f(x) = 2^x$ $\qquad f(4) = 2^4 = 16$

$\log_a a$

$\boxed{f(x) = 2^x = 16}$ $\qquad 2^x = 2^4$

$\Rightarrow x = 4$

$\Rightarrow \log_2 2^x = \log_2 16 \Rightarrow x \log_2 2 = \log_2 2^4$

$\Rightarrow x \cdot 1 = 4 \times 1$

$g(x) = \log_2 x$ $\qquad \Rightarrow x = 4$

$f \circ g = f(g(x)) = f(\log_2 x)$

$= 2^{\log_2 x} = x$

$f(x) = a^x \quad 0<a<1, \quad a>1$

$\boxed{g(x) = \log_a x \qquad 0<a<1, \quad a>1}$ $\qquad g(5) = \log_a 5$

$= \log_2 5$

① $\boxed{\log_a bc = \log_a b + \log_a c}$

$\log_a b = m$

$\Rightarrow a^m = b$

$a^n = c \Leftrightarrow \boxed{\log_a c = n}$

L.H.S $\log_a bc = m$

$\Rightarrow \boxed{a^m = bc}$

R.H.S $\log_a b + \log_a c = m$

taking $\Rightarrow a^{(\log_a b + \log_a c)} = a^m$

$\Rightarrow a^{\log_a b} \times a^{\log_a c} = a^m$

$\Rightarrow b \times c = a^m$

$\Rightarrow \boxed{bc = a^m}$

$\boxed{\log_a bc = }$

$\Leftrightarrow \boxed{a^x = bc}$

//

$\Rightarrow$

$\boxed{a^{m+n} = a^m \cdot a^n}$

$\boxed{a^{\log_a b} = b}$

$\Downarrow$

$\boxed{a^{\log_a b} = n = b}$

$\Rightarrow \log_a a^{\log_a b} = n$

$\Rightarrow \log_a b (\log_a a) = \log_a n$

$\Rightarrow \log_a b = \log_a n$

$\Rightarrow n = b$

(ii) $\log_a \frac{m}{n} = \log_a m - \log_a n$

(iii) $\boxed{a^{\log_a b} = b}$

$\log_{x^5} b + \log_{y^5} c = 0$

(iv) $\boxed{\log_{a^n} b = \frac{1}{n} \log_a b}$

$\log_{32} c + \log_{16} b$

$2^5 \quad 2^4$

(v) $\boxed{\log_a b^n = n \log_a b}$

$\Downarrow$

(vi) $\log_a 1/b = -\log_a b$ //

$1/b = (b)^{-1}$

$\boxed{\log_a bc = \log_a b + \log_a c}$

$\log_a b + \log_a c = m$

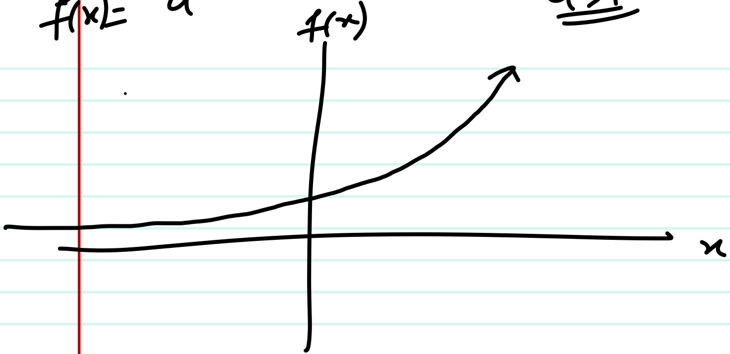
$\Rightarrow a^{(\log_a b + \log_a c)} = a^m$

$\Rightarrow a^{\log_a b} \times a^{\log_a c} = a^m$

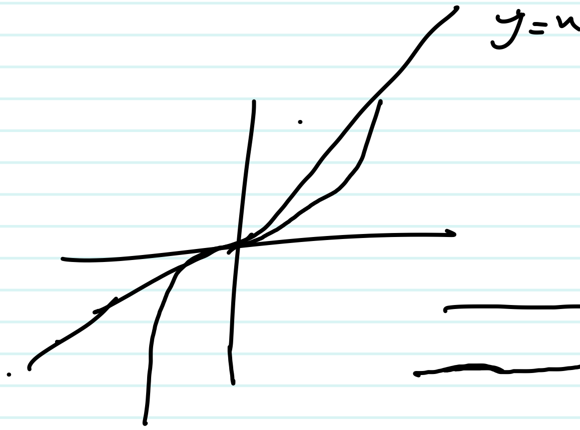
$\Rightarrow b \times c = a^m$

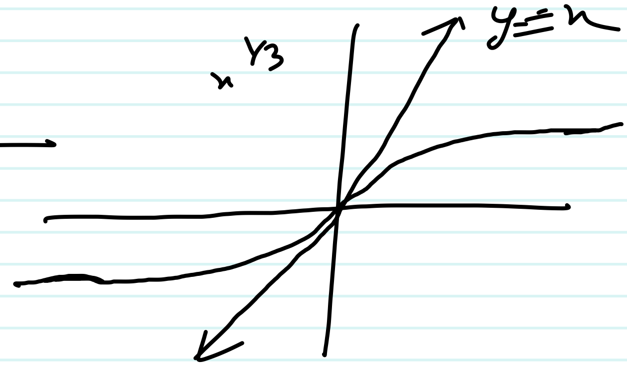
$\Rightarrow a^m = bc \Leftrightarrow \boxed{\log_a bc = m}$

$f(x) = a^x$ $\quad \underline{\underline{a>1}}$



$\underline{\underline{f(x) = x^3}}$



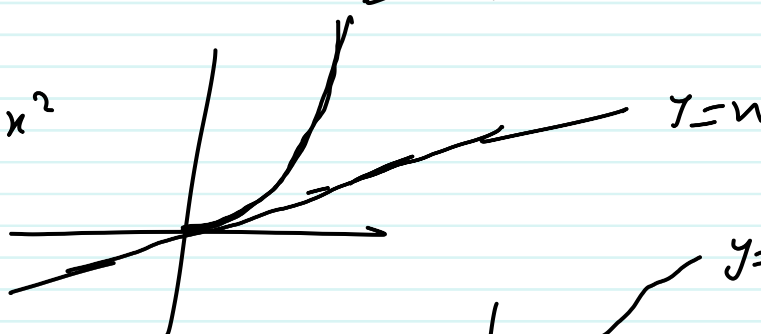


$\underline{g(x) = x^{1/3}}$

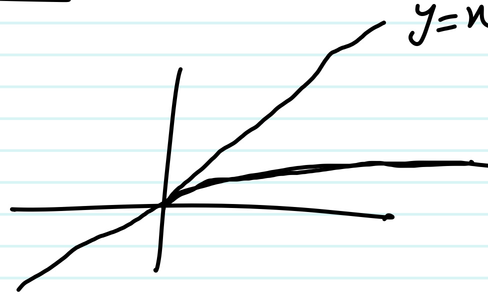
$$\begin{aligned} f \circ g &= f(g(x)) \\ &= f(x^{1/3}) \\ &= (x^{1/3})^3 \\ &= x^{\frac{1}{3} \times 3} \\ &= x \end{aligned}$$

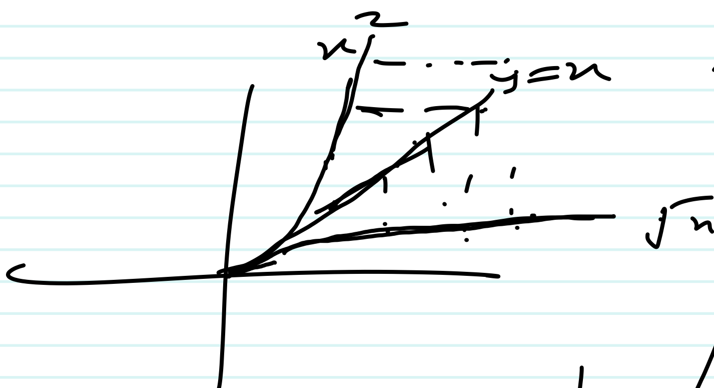
$f(x) = x^2$

$\underline{x \in [0, \infty)}$



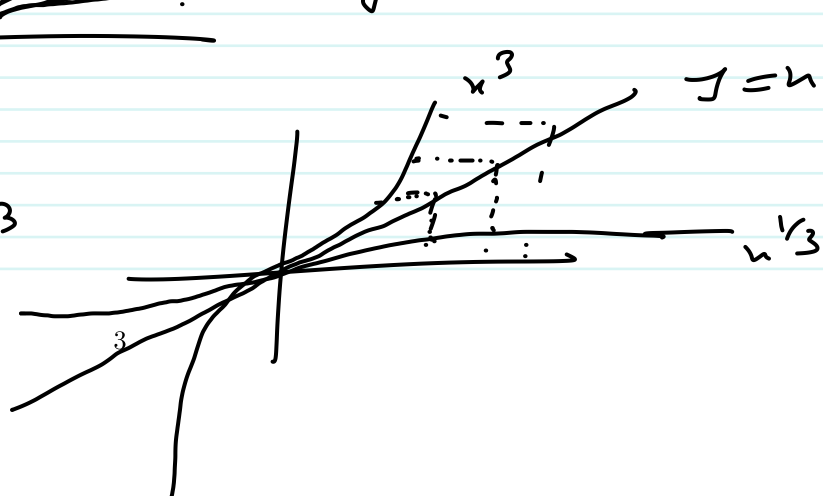
$g(x) = \underline{\underline{\sqrt{x}}}$





$f(x) = x^3$

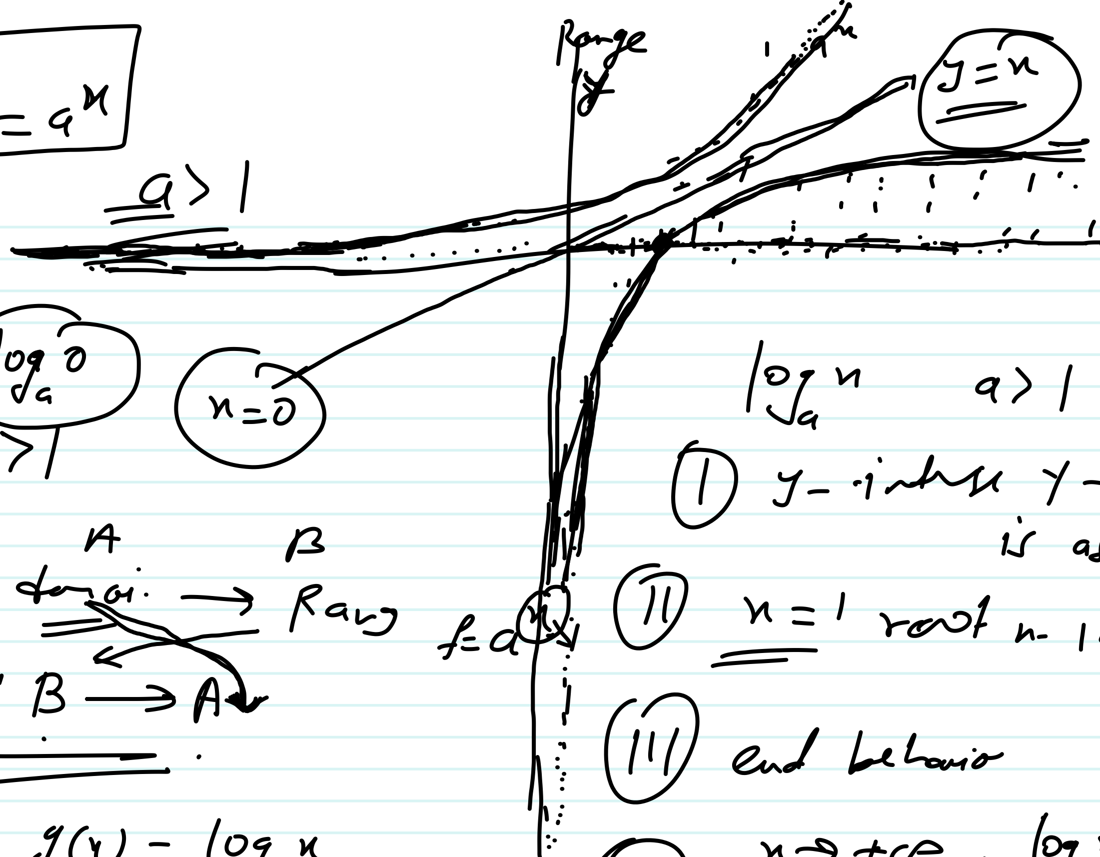
$g(x) = x^{1/3}$



$$\boxed{f(x) = a^x}$$

$a > 1$

$g(x) = \log_a(x)$



$\log_a 0$

$f(x) = a^x,\ a > 1$

$f(x) \to \quad f: \text{domain } A \to \text{Range } B$

$g(x) \to \quad g: B \to A$

$\log_a x \quad a > 1$

(I) y-intercept y-axis is asympt

(II) $x = 1$ root x-inter

(III) end behavior

(IV) $x \to +\infty,\ \log_a x \to +\infty$

$\boxed{x \to 0^+,\ \log_a x \to -\infty}$

(V) increasing

(I) One to one

$g(x) = \log_a x$

$g(a) = \log_a a = 1$

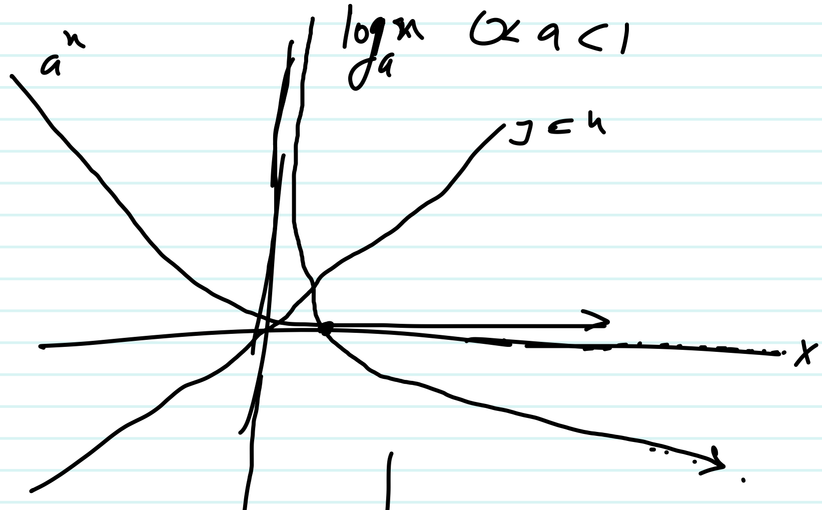
$= \boxed{\log_a 1 = 0}$

$a^0 = 1$

$\Rightarrow \boxed{\log_a 1 = 0}$

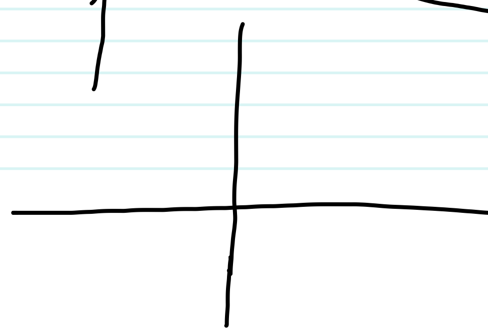
$\log_a x \quad a > 1 \qquad a^x$

$\log_a x \quad 0 < a < 1$



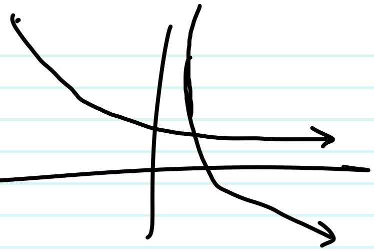
$a^x \qquad 0 < a < 1$

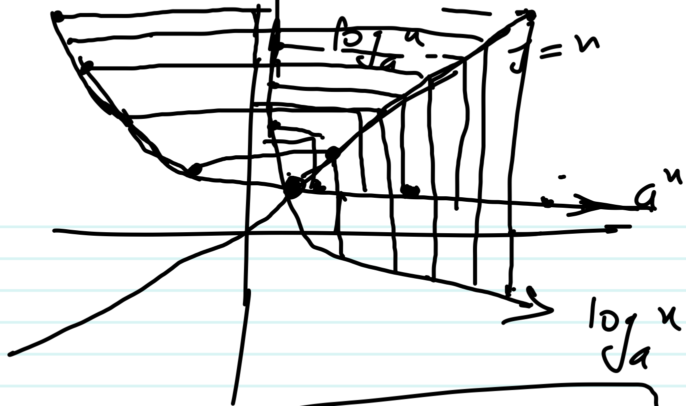
$y = \log_a x \qquad 0 < a < 1$



$\log_a x \quad 0 < a < 1$

a^x





$\log x = \log_{10} x$

$\ln x = \log_e$

$$\boxed{f(x) = \log_{10}(e^x - e^{-x})}$$

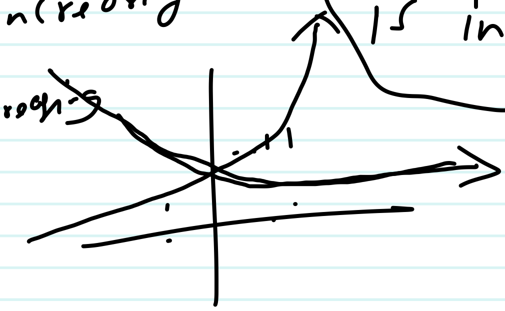
- is one-one
- is increasing
- is decreasing
- is invertible

$\log(a - b) \neq \log a - \log b$

$\log m/n = \log m - \log n$

e^x increasing

e^{-x} decreasing



e^{-1}

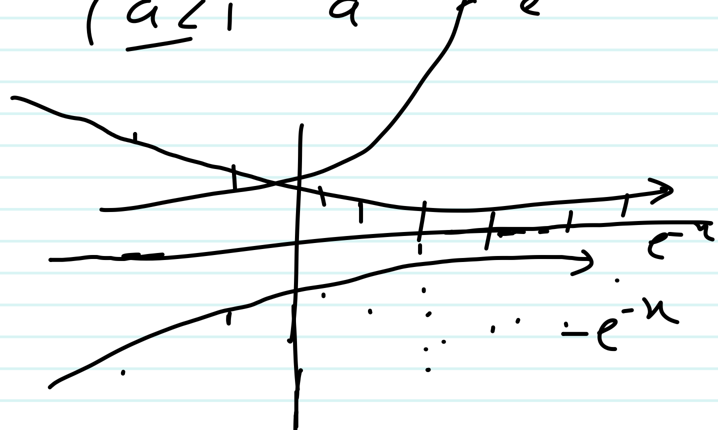
$\frac{1}{e^x}$

$\frac{1}{x} = 1$

$\left(\frac{1}{e}\right)^x$

$a > 1 \quad a^x$

$1 a < 1 \quad a^x \quad e^x$



e^x increas

e^{-x} decreas.

$-e^{-x}$ increasing

$e^x - e^{-x}$ increasing e^{-x}

$\log_{10}(e^x - e^{-x})$ increas.

$g(x)$

$10 > 1$

$x = 3, \quad g(3) = e^3 - e^{-3}$

$= x = 5 \quad g(5) = e^5 - e^{-5}$

$\log(e^x - e^{-x})$ $\quad (0, \infty)$

$g(3) < g(5)$

$f(3) = \log(e^3 - e^{-3})$

$f(5) = \log(e^5 - e^{-5})$

$\boxed{f(3) < f(5)}$

$(0, \infty)$ $a>0$ $(0, \infty)$

(a, ∞) $a>0$

$\boxed{e^x - e^{-x} > 0}$

$e^x > e^{-x}$

$\Rightarrow x > -x$

$\Rightarrow 2x > 0$

$\Rightarrow x > 0$

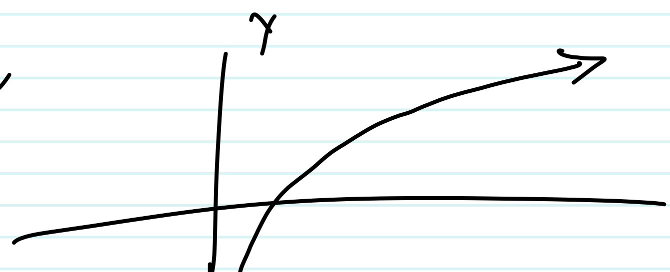
$(0, \infty)$

$a^b = a^c \Rightarrow b = c$

$\log x \quad (0, \infty)$

$\boxed{\log_{10} 5} =$

$\log_a x \quad a > 1$



$f(x) = \log\left((\log x)^2 - 8\log x + 15\right)$

$(\log x)^2 - 8\log x + 15 > 0 \qquad \log x = y$

$\boxed{y^2 - 8y + 15} = (y-3)(y-5)$

$3+5$

$y^2 - 8y + 15 > 0$

$y > 5 \rightarrow$

$y > 3, \; y > 5 \Rightarrow y > 5$

$y = 4$

$\_\_\_ < 0$ $\quad (3,5)$ $\quad [3,5]$ 4

$\_\_\_ > 0$ $\quad y < 3 \Rightarrow$ $\quad y < 3$

$y = 3$ $\quad (y-3)(y-5) = 0$

$y = \log x$ $\quad$ when $y = 3,\ y = 5$

$y > 5$

$\log x > 5$ $\quad y < 3$ $\quad a^5 = m$ $\quad y < 3$

$\Updownarrow$

$\Rightarrow 10^5 > x$ $\quad \log x < 3$ $\quad \log_a m = 5$

$x < 10^5$ $\quad \Rightarrow 10^3 < x$

$y - 3 = 0$

$y - 5 = 0$

$\log_{10} x > 5$

$y > 5$

$y < 3 \Rightarrow$ $\quad 10^{\log_{10} x} > 10^5$ $\quad a > 1$

$3 < 5$

$2^3 < 2^5$

$8 \leq 32$

$(0, 3) \cup (5, \infty)$ $\quad x > 10^5$

$\Downarrow$

$\log_{10} x = m$

$\Rightarrow 10^m = x$

$(0, 10^3) \cup (10^5, \infty)$